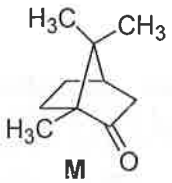
PART II : CHEMISTRY

SECTION 1 (Maximum Marks: 32)

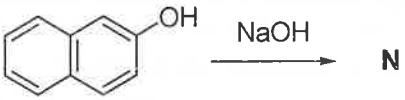
- This section contains **EIGHT** questions
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme:
  +4 If the bubble corresponding to the answer is darkened
  0 In all other cases

Q.21 The total number of stereoisomers that can exist for **M** is



M

Q.22 The number of resonance structures for **N** is



N

Q.23 The total number of lone pairs of electrons in N_2O_3 is

Q.24 For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin-only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is
[Atomic number of Fe = 26]

Answers for the above questions

Ans for Q.21: 2	**Ans for Q.22: 9**	**Ans for Q.23: 8**	**Ans for Q.24: 4**

Q.25 Among the triatomic molecules/ions, $BeCl_2$, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^- and XeF_2, the total number of linear molecule(s)/ion(s) where the hybridization of the central atom does not have contribution from the *d*-orbital(s) is
[Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54]

Q.26 Not considering the electronic spin, the degeneracy of the second excited state ($n = 3$) of H atom is 9, while the degeneracy of the second excited state of H^- is

Q.27 All the energy released from the reaction $\mathbf{X} \rightarrow \mathbf{Y}$, $\Delta_r G^0 = -193\,\text{kJ mol}^{-1}$ is used for oxidizing $\mathbf{M}^+$ as $\mathbf{M}^+ \rightarrow \mathbf{M}^{3+} + 2e^-$, $E^0 = -0.25\,V$.
Under standard conditions, the number of moles of $\mathbf{M}^+$ oxidized when **one** mole of **X** is converted to **Y** is
[$F = 96500\ \text{C mol}^{-1}$]

Q.28 If the freezing point of a 0.01 molal aqueous solution of a cobalt(III) chloride-ammonia complex (which behaves as a strong electrolyte) is $-0.0558\,°\text{C}$, the number of chloride(s) in the coordination sphere of the complex is
[K_f of water = 1.86 K kg mol^{-1}]

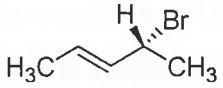
Answers for the above questions

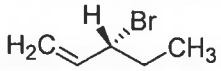
Ans for Q.25: 4	**Ans for Q.26: 3**	**Ans for Q.27: 4**	**Ans for Q.28: 1**

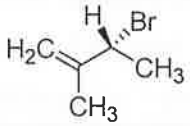
SECTION 2 (Maximum Marks: 40)

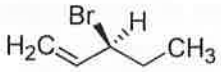
- This section contains **TEN** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme:
  - +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
  - 0 If none of the bubbles is darkened
  - –2 In all other cases

Q.29 Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is(are)

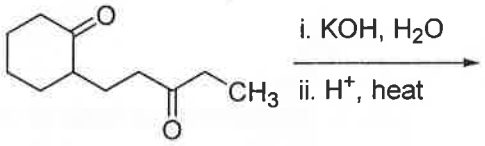
(A) 

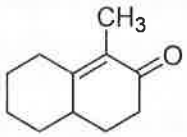
(B) 

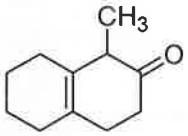
(C) 

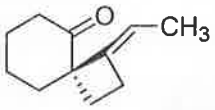
(D) 

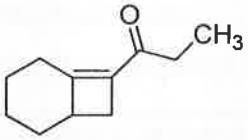
Q.30 The major product of the following reaction is



(A) 

(B) 

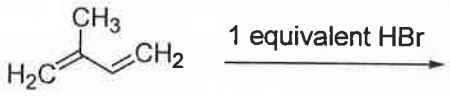
(C) 

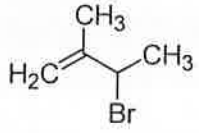
(D) 

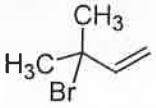
Answers for the above questions

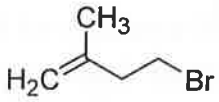
Ans for Q.29: (B) and (D)	Ans for Q.30: (A)

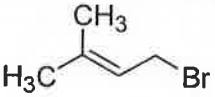
Q.31 In the following reaction, the major product is



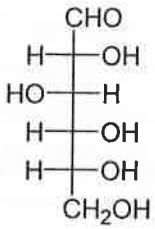
(A) 

(B) 

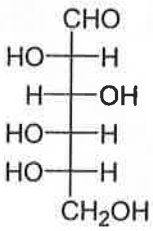
(C) 

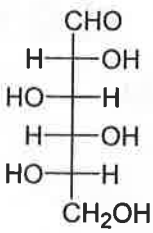
(D) 

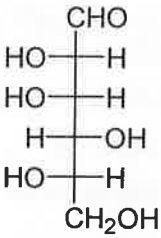
Q.32 The structure of D-(+)-glucose is

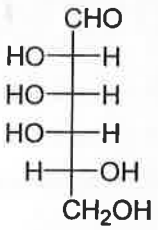


The structure of L-(–)-glucose is

(A) 

(B) 

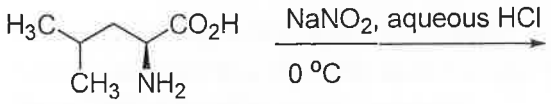
(C) 

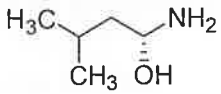
(D) 

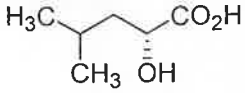
Answers for the above questions

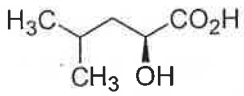
Ans for Q.31: (D)	Ans for Q.32: (A)

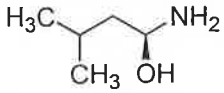
Q.33 The major product of the reaction is



(A) 

(B) 

(C) 

(D) 

Q.34 The correct statement(s) about Cr^{2+} and Mn^{3+} is(are)
[Atomic numbers of Cr = 24 and Mn = 25]

(A) Cr^{2+} is a reducing agent

(B) Mn^{3+} is an oxidizing agent

(C) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration

(D) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration

Q.35 Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is (are)

(A) Impure Cu strip is used as cathode

(B) Acidified aqueous $CuSO_4$ is used as electrolyte

(C) Pure Cu deposits at cathode

(D) Impurities settle as anode-mud

Q.36 Fe^{3+} is reduced to Fe^{2+} by using

(A) H_2O_2 in presence of NaOH

(B) Na_2O_2 in water

(C) H_2O_2 in presence of H_2SO_4

(D) Na_2O_2 in presence of H_2SO_4

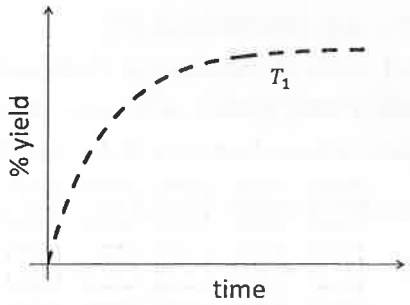
Answers for the above questions

Ans for Q.33: (C)	**Ans for Q.34: (A), (B) and (C)**
Ans for Q.35: (B), (C) and (D)	**Ans for Q.36: (A) and (B)**

Q.37 The %yield of ammonia as a function of time in the reaction

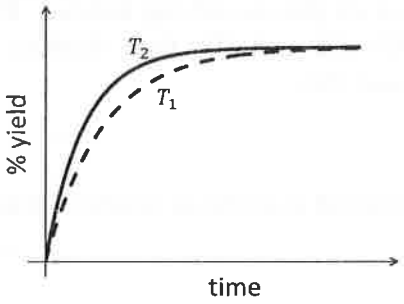
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H < 0$

at (P, T_1) is given below.

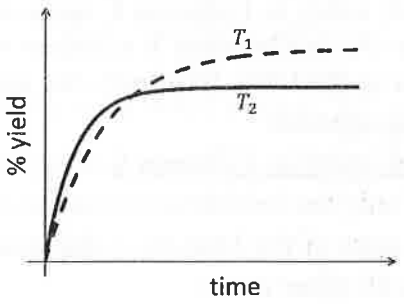


If this reaction is conducted at (P, T_2), with $T_2 > T_1$, the %yield of ammonia as a function of time is represented by

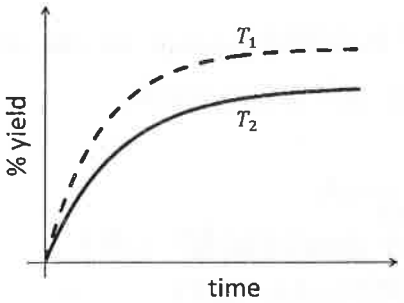
(A)



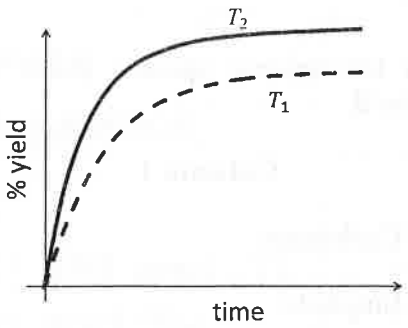
(B)



(C)



(D)



Q.38 If the unit cell of a mineral has cubic close packed (*ccp*) array of oxygen atoms with ***m*** fraction of octahedral holes occupied by aluminium ions and ***n*** fraction of tetrahedral holes occupied by magnesium ions, ***m*** and ***n***, respectively, are

(A) $\frac{1}{2}, \frac{1}{8}$ (B) $1, \frac{1}{4}$ (C) $\frac{1}{2}, \frac{1}{2}$ (D) $\frac{1}{4}, \frac{1}{8}$

Answers for the above questions

Ans for Q.37: (B)	Ans for Q.38: (A)

SECTION 3 (Maximum Marks: 16)

- This section contains **TWO** questions
- Each question contains two columns, **Column I** and **Column II**
- **Column I** has **four** entries (A), (B), (C) and (D)
- **Column II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column I** with the entries in **Column II**
- One or more entries in **Column I** may match with one or more entries in **Column II**
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below:

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

- For each entry in **Column I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- Marking scheme:

  For each entry in **Column I**,

  +2 If only the bubble(s) corresponding to all the correct match(es) is(are) darkened
  0 If none of the bubbles is darkened
  –1 In all other cases

Q.39 Match the anionic species given in Column I that are present in the ore(s) given in Column II.

	Column I		Column II
(A)	Carbonate	(P)	Siderite
(B)	Sulphide	(Q)	Malachite
(C)	Hydroxide	(R)	Bauxite
(D)	Oxide	(S)	Calamine
		(T)	Argentite

Answers for the above question

Ans for Q.39:
(A) Matches to (P), (Q) and (S)
(B) Matches to (T)
(C) Matches to (Q) and (R)
(D) Matches to (R)

Q.40 Match the thermodynamic processes given under Column I with the expressions given under Column II.

	Column I		Column II
(A)	Freezing of water at 273 K and 1 atm	(P)	$q = 0$
(B)	Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions	(Q)	$w = 0$
(C)	Mixing of equal volumes of two ideal gases at constant temperature and pressure in an isolated container	(R)	$\Delta S_{sys} < 0$
(D)	Reversible heating of $H_2(g)$ at 1 atm from 300 K to 600 K, followed by reversible cooling to 300 K at 1 atm	(S)	$\Delta U = 0$
		(T)	$\Delta G = 0$

END OF PART II : CHEMISTRY

Answers for the above question

Ans for Q.40:
(A) Matches to (R) and (T)
(B) Matches to (P), (Q) and (S)
(C) Matches to (P), (Q) and (S)
(D) Matches to (P), (Q), (S) and (T)